

[Chathurya V R 251801045-2025-AIDS](#) ▾**C2****Started on** Thursday, 27 August 2026, 8:51 AM**State** Finished**Completed on** Thursday, 17 September 2026, 9:41 AM**Time taken** 21 days**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  // Divide and Conquer function to find the floor of x
4  int findFloor(int nums[], int low, int high, int x) {
5      // Base case: If search space is exhausted
6      if (low > high) {
7          return -1;
8      }
9
10     // If element is greater than or equal to all elements in the current range
11     if (x >= nums[high]) {
12         return nums[high];
13     }
14
15     // Divide step: find the middle index
16     int mid = low + (high - low) / 2;
17
18     // If exact match found
19     if (nums[mid] == x) {
20         return nums[mid];
21     }
22
23     // If x is smaller than nums[mid], the floor must lie in the left half
24     if (x < nums[mid]) {
25         return findFloor(nums, low, mid - 1, x);
26     }
27
28     // If x is greater than nums[mid], nums[mid] is a candidate.
29     // Check if a better floor exists in the right half.
30     int rightFloor = findFloor(nums, mid + 1, high, x);
31
32     if (rightFloor != -1) {
33         return rightFloor;
34     } else {
35         return nums[mid];
36     }
37 }
38
39 int main() {
40     int n;
41     scanf("%d", &n);
42
43     int nums[n];
44     for (int i = 0; i < n; i++) {
45         scanf("%d", &nums[i]);
46     }
47
48     int x;
49     scanf("%d", &x);
50
51     int result = findFloor(nums, 0, n - 1, x);
52

```

	Input	Expected	Got	
✓	6	2	2	✓
	1			
	2			
	8			
	10			
	12			
	19			
	5			
✓	5	85	85	✓
	10			
	22			
	85			
	108			
	129			
	100			
✓	7	9	9	✓
	3			
	5			
	7			
	9			
	11			
	13			
	15			
	10			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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